

Question Bank:

Class-1

Find the Fourier Transform of the following:

$$1) \quad f(t) = \begin{cases} a & -l < t < 0 \\ b & 0 < t < l \end{cases} \quad \text{Solution: } F[f(t)] = \frac{i}{\omega} \left[(a-b) - ae^{i\omega l} + be^{-i\omega l} \right]$$

$$2) \quad f(t) = e^t, \text{ where } |t| < a \quad \text{Solution: } F[f(t)] = \frac{2 \sinh(i\omega - 1)a}{i\omega - 1}$$

$$3) \quad f(t) = \begin{cases} 1-t, & 0 < t < 1 \\ 1+t, & -1 < t < 0 \\ 0, & |t| > 1 \end{cases} \quad \text{Solution: } F[f(t)] = \frac{2(1 - \cos \omega)}{\omega^2}$$

Find the Inverse Fourier Transform of the following:

$$4) \quad \frac{e^{-i\omega}}{2(1+i\omega)} \quad \text{Solution: } F^{-1}[F(s)] = \frac{1}{2} e^{-(t-1)} U(t-1)$$

$$5) \quad \frac{e^{-2i\omega}}{(2+3i\omega)} \quad \text{Solution: } F^{-1}[F(s)] = \frac{1}{3} e^{\frac{-2(t-2)}{3}} U(t-2)$$

$$6) \quad \frac{e^{4i\omega}}{(3+i\omega)} \quad \text{Solution: } F^{-1}[F(s)] = e^{-3(t-4)} U(t-4)$$